

AI/ML-Based Personalized Diet Plan Generator from Medical Reports

Milestone 3: NLP / AI-Based Text Interpretation

Milestone Objectives

The objective of Milestone 3 was to enable the system to understand **unstructured textual information** present in doctor prescriptions and medical notes and convert it into **actionable diet recommendations**.

As per the project plan, this milestone aimed to:

- Integrate NLP / AI techniques to interpret doctor notes and prescriptions
- Extract meaningful dietary instructions from medical text
- Map extracted text to actionable diet guidelines
- Achieve at least **80% conversion** of textual notes into usable diet rules

Problem Context

Medical prescriptions and doctor notes often contain critical dietary advice such as:

- “Avoid sugar”
- “Reduce carbohydrate intake”
- “No fried food”
- “Increase fiber and vegetables”

However, these instructions are typically:

- Handwritten or scanned
- Difficult for patients to interpret
- Not structured for automated processing

After completing **Milestone 1 (OCR & preprocessing)** and **Milestone 2 (ML-based numeric health analysis)**, the system required an **NLP layer** to interpret unstructured medical text and bridge the gap between raw doctor advice and personalized diet planning.

System Overview (Milestone 3)

In this milestone, the system workflow was extended as follows:

1. User uploads a **doctor prescription image** through the frontend UI
2. Backend performs **OCR** to extract textual content
3. NLP module interprets the extracted text

4. Dietary instructions are mapped to **structured diet guidelines**
5. Results are displayed to the user in a clear and understandable format

Prescription Upload & OCR Integration

The frontend allows users to upload prescription images in **JPEG or PNG format**. The backend processes the uploaded file and applies **Tesseract OCR with OpenCV preprocessing**.

OCR Output Includes:

- Doctor instructions
- Short advisory notes
- Medication-related comments (when present)

This step reuses and extends the OCR pipeline implemented in **Milestone 1**.

NLP-Based Text Interpretation Approach

A **hybrid NLP approach** was adopted to ensure reliability and explainability.

Text Preprocessing

- Converted extracted text to lowercase
- Removed special characters and noise
- Normalized OCR output for consistency

Rule-Based NLP Extraction

A predefined set of medically relevant dietary keywords and phrases was used to identify actionable instructions.

Examples of Extracted Rules:

- Avoid sugar
- Low carbohydrate diet
- Increase fiber intake
- Avoid fried or oily food
- Increase vegetable consumption

This rule-based approach ensures stable performance even when OCR output quality varies.

Mapping to Actionable Diet Guidelines

Each extracted rule is mapped to a user-friendly and practical diet recommendation.

Extracted Rule	Actionable Diet Guideline
Avoid sugar	Avoid sweets, sugary drinks, and desserts
Low carb diet	Reduce rice, bread, and refined carbohydrates
Increase fiber	Consume fruits, vegetables, and whole grains
Avoid fried food	Avoid deep-fried and oily foods

This process converts raw medical advice into **practical daily diet actions**.

Personalized Meal Plan Generation (Frontend Output)

As part of Milestone 3, the system was extended beyond text interpretation to generate **structured and user-friendly meal plans** based on the extracted medical information and dietary rules.

After OCR and NLP processing, the frontend dynamically generates a **personalized daily meal plan** divided into:

- **Breakfast**
- **Lunch**
- **Dinner**
- **Recommended Snacks**

Each meal card includes:

- Meal name and description
- Calorie information
- Macronutrient indicators (carbohydrates, protein, fat)

This structured presentation helps users clearly understand **what to eat at different times of the day**, rather than receiving generic dietary advice.

Additional Features

- Export meal plan as **PDF**
- Export meal plan as **JSON**

These features improve usability, support offline access, and enable future integration with nutrition or health tracking systems.

The inclusion of breakfast, lunch, and dinner ensures that interpreted medical advice is translated into **practical, actionable daily eating habits**, completing the end-to-end pipeline from prescription analysis to meal planning.

Upload Doctor Prescription

Choose File 10.jpg

Analyze Prescription

Extracted Medical Information

Health Conditions Detected:

General health maintenance

Dietary Restrictions:

- None specified

Your Personalized Meal Plan

Export PDF

Export JSON

BREAKFAST

Whole grain cereal with milk

High-fiber cereal with low-fat milk and banana

300 cal

12g

50g

LUNCH

Turkey sandwich with vegetables

Whole grain bread with turkey, lettuce, tomato

420 cal

28g

45g

DINNER

Vegetable curry with brown rice

Mixed vegetable curry with aromatic brown rice

420 cal

12g

70g

Recommended Snacks

Handful of nuts

Almonds, walnuts, or mixed nuts

160 cal

Vegetable sticks with hummus

Carrots, celery, cucumber with hummus

100 cal

Key Learnings

Through Milestone 3, the following concepts and skills were developed:

- End-to-end OCR and NLP pipeline design
- Handling noisy real-world medical text
- Rule-based NLP for explainable healthcare AI
- Integration of AI pipelines with frontend UI
- Designing user-centric healthcare applications

Conclusion

Milestone 3 successfully enabled the system to interpret unstructured doctor prescriptions using NLP techniques and convert them into actionable diet recommendations. By combining OCR, text preprocessing, and rule-based NLP, the system achieved reliable performance and met the target of converting **over 80% of textual medical notes** into meaningful dietary guidance.

This milestone completes the **text understanding layer** of the AI Diet Planner and prepares the system for final personalization and advanced diet recommendation features in subsequent stages.

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